

Homework 1

Data Analysis and Machine Learning with Python

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Dataset 1: housing_data.csv Field Descriptions:

'Area': House area,

'No. of Rooms': Number of rooms (not including Bathrooms), 'No. of

Bathrooms': Number of bathrooms,

'Location': Location of the house (Suburb, City Center, or Rural),

'Miles (dist. between NTU and house)': Distance between the house and the NTU,

'Rent Price per Month': Monthly rent price, 'Sell

Price': House selling price

I. Part 1: Questions Related to Data Analysis

Q1. What steps will you take upon receiving this dataset before commencing data analysis?

會先認識資料並整理資料。

(一)整理資料

檢查是否有缺失值 (NaN)、檢查是否有重複數據、檢查是否有負數、檢查是否有不合理的數據類型

```
In [3]: # 檢查缺失值
missing_values = data.isnull().sum()
print("缺失值:\n", missing_values)
```

```
缺失值:
Area                                0
No. of Rooms                        0
No. of Bathrooms                    0
Location                            0
Miles (dist. between school and house) 0
Rent Price per Month                0
Sell Price                          0
dtype: int64
```

```
In [4]: # 檢查重複值
duplicate_rows = data.duplicated().sum()
print("\n重複行數:", duplicate_rows)
```

```
重複行數: 0
```

```
In [5]: # 檢查是否有負數 (僅針對數值型欄位)
negative_values = (data.select_dtypes(include=['number']) < 0).sum()
print("\n負數值:\n", negative_values)
```

```
負數值:
Area                                0
No. of Rooms                        0
No. of Bathrooms                    0
Miles (dist. between school and house) 0
Rent Price per Month                0
Sell Price                          0
dtype: int64
```

(二)認識資料

在HW1-1中，`data.info()`我使用了去了解資料的欄位。`data.describe()`去了解資料的大致分布情形，分布資料可以讓我後續在篩選的時候知道自己訂下的條件可能符合的項目可能有多少。`data.head()`去大致看一下前面五筆資料。

比較特別的是，我還使用`avg_sell_price_by_location = data.groupby("Location")["Sell Price"].mean().reset_index()`與`avg_sell_price_by_rooms = data.groupby("No. of Rooms")["Sell Price"].mean().reset_index()`去了解Location和No. of Rooms對Sell Price的相關性。

Q2. If you inquire about Q1 from an AI tool, what responses will you receive? Do you find them reasonable? If not, how will you rectify it?

AI(ChatGPT)跟我說要load the dataset, understand the dataset, handle missing data, check for duplicates, check data types and convert if

necessary, handle outliers, normalize or standardize data, perform EDA, feature engineering, save cleaned dataset 我覺得AI說得合理，基本上用比我更精準的詞彙將資料分析前的處理展開來說，也與課程中所教的內容相符合。feature engineering是我不熟悉的詞彙，透過ChatGPT的回答才了解的。但也或許我說它說的合理是因為我的知識儲備量並不足夠道能判斷所有它所說的詞彙。

Q3. If you are restricted to renting a house, which one or ones will you select, and why?

我是假設自己是對租房品質有一定要求但預算又有限的租屋客。

```
filtered_data = data[(data['No. of Rooms'] >= 2) & (data['Miles (dist. between school and house)'] <= 100)&(data['Rent Price per Month'] < 10000)&(data['Location'] == 'City Center')]
```

可以幫我篩選滿足房間數大於2、學校與房子距離小於100、每個月租金小於一萬且位於市中心的房子。這樣確保所有房子都是我能接受的居住條件。

```
min_rent_data = filtered_data.loc[filtered_data['Rent Price per Month'].idxmin()]
```

然後我再選擇其中價格最低的租，篩出來的是524號。

Area	1655
No. of Rooms	2
No. of Bathrooms	1
Location	City Center
Miles (dist. between school and house)	60
Rent Price per Month	6451
Sell Price	49151508
Name: 524, dtype: object	

Q4. Assuming you have enough funds to purchase a house, will you opt to continue renting or proceed with a purchase? If renting, which one will you choose? If buying, which one will you select? Why?

利用Q3已經找到的數據，再進行以下處理。

計算 Sell Price / Rent Price per Month 的比值

```
filtered_data['Sell Price to Rent Ratio'] = filtered_data['Sell Price'] / filtered_data['Rent Price per Month']
```

(1)選取比值最大的

```
max_rent_data = filtered_data.loc[filtered_data['Sell Price to Rent Ratio'].idxmax()] max_rent_data
```

比值最大者表示要達到買到需要付最多個月房租才能買到這間房子，因此我會選它租，被篩選出的是524。

Area	1655
No. of Rooms	2
No. of Bathrooms	1
Location	City Center
Miles (dist. between school and house)	60
Rent Price per Month	6451
Sell Price	49151508
Sell Price to Rent Ratio	7619.207565

Name: 524, dtype: object

(2) 選取比值最小的

```
min_rent_data = filtered_data.loc[filtered_data['Sell Price to Rent Ratio'].idxmin()]
```

```
min_rent_data
```

比值最小者表示要達到買到需要付最少個月房租就能買到這間房子，因此我會選它買，被篩選出的是996。

Area	1262
No. of Rooms	2
No. of Bathrooms	1
Location	City Center
Miles (dist. between school and house)	38
Rent Price per Month	9904
Sell Price	20223887
Sell Price to Rent Ratio	2041.991821

Name: 996, dtype: object

Q5. Are there any properties with rent or selling prices that seem unusually high or low? Why?

(1) 想法一:使用IQR分析後，可以把異常值挑出來

但看完篩選出的數據依然有近千筆，無法看出趨勢。

(概念是詢問AI的)

	Area	No. of Rooms	No. of Bathrooms	Location	Miles (dist. between school and house)	Rent Price per Month	Sell Price	is_outlier
0	1360	1	1	Rural	463	7401	74446632	False
1	1794	3	1	Suburb	210	9259	76199794	False
2	1630	2	1	Suburb	157	16469	16249579	False
3	1595	1	1	Suburb	133	18096	24291317	False
4	2138	1	1	Suburb	10	9923	50273384	False
...	...	...	...	...	...	...	...	...
995	1375	1	1	City Center	33	16039	8670492	False
996	1262	2	1	City Center	38	9904	20223887	False
997	1702	3	1	Rural	180	13260	76086163	False
998	1705	3	1	Rural	418	6504	38329598	False
999	1163	1	1	Rural	237	15813	68023737	False

1000 rows × 8 columns

(2)想法二:以Rent Price per Month為y軸，Sell Price為x軸看每筆資料得分佈

使用散佈圖可以發現資料的分布實際上沒有線性規律。因此推測此份資料可能是隨機產生的。



II. Part 2: Questions Related to the Use of AI Tools:

A. Did you use AI tools to assist in completing Q3 to Q5?

Yes

B1. Which AI tool(s) did you use?

ChatGPT

B2. With so many AI tools available, why did you choose this particular AI tool (or these AI tools) for assistance?

因為我認為ChatGPT在提供想法方面很方便，其他ai像是 copilot、cursor、Claude 我會比較偏向輪流用來查代碼撰寫。

B3. What factors did you consider when selecting the AI tool? e.g., functionality, learning curve, accuracy, data security, etc.

功能性跟回答的簡潔程度都有影響吧

B4. Did you compare the results of multiple AI tools? If so, what were the differences between the results, and what do you think caused these differences?

沒有欸，執行了ChatGPT給我的想法一後，我自己就有了想法二。

B5. How did you ensure that the output from the AI tool(s) was reliable?

我是覺得可以用excel直接來找IQR驗證，但因為我後來沒有繼續延續這代碼，所以就沒有再進一步地確認。

B6. How did you design your prompt (command or question) to address Q3 to Q5? Did you adjust or iterate on this prompt?

這是我的房屋資料，請給我回答資料分析問題: Are there any properties with rent or selling prices that seem unusually high or low? Why?的思路我應該用python 寫什麼

與ChatGPT對話連結:<https://chatgpt.com/share/67d553a7-5524-800d-a817->

852f0837fc62

這是我打給AI的提示詞，我是將題目加上一些限制去尋問它。

B7. Was your prompt directly copied from the assignment question, or did you modify it based on the assignment requirements? What specific modifications did you make?

我是將作業題目加上一些限制去尋問它，希望他能根據csv檔以Python回答，然後多給我一些思路想法。

B8. When solving Q3 to Q5, how did you break down a large problem into smaller, manageable parts? Can you provide an example?

因為我只是問Q5的初步想法所以沒有遇到這個問題

B9. How did you plan the steps to solve the Q3 to Q5? Was there a logical order to these steps?

因為我只是問Q5的初步想法所以沒有遇到這個問題

B10. During the process of completing Q3 to Q5, did you encounter any challenges? How did you overcome them? How did the AI tool(s) help in this regard?

我是在Q5的初步想法上遇到了困難，ChatGPT給我的想法有刺激到我後面會去畫散佈圖。

B11. After using the AI tool(s) to complete Q3 to Q5, did you learn any new data analysis techniques or knowledge?

我覺得沒有，因為相關觀念老師上課都有講過。

B12. Throughout the process of solving Q3 to Q5, what aspects do you think AI tools cannot fully replace human thinking?

AI只是分析了資料，但對於資料無明顯線性分布還是需要我自己判斷，因此我認為視情況下結論是人類不能被AI取代的能力。

Dataset 2: family_data.csv

Field Descriptions:

'Family': Family ID,

'Member': Member (Adult or Child), 'Income':

Annual income,

'Spend': Annual spending

I. Part 1: Questions Related to Data Analysis

Q1. Which family boasts the highest annual income, and which has the lowest? How do you ascertain this?

family 6有最高的income，family 94有最低的income。

```
data_grouped = data.groupby("Family").agg({"Income": "sum",  
"Spend": "sum"}) 這條程式幫助我以家庭為單位將各家的收入與支出進行加總
```

```
highest_income_family = data_grouped[data_grouped["Income"] ==  
data_grouped["Income"].max()]與lowest_income_family =  
data_grouped[data_grouped["Income"] ==  
data_grouped["Income"].min()]這兩條式子幫我篩出了收入最高與最低的家  
庭，因此我找到family 6有最高的income，family 94有最低的income。
```

Q2. Which families do not possess adequate annual income to cover all members' spending? What is the maximum shortfall? How do you determine this?

1. 不存在入不敷出的家庭。

```
2. families_income_less_than_spend =  
data_grouped[data_grouped["Income"] < data_grouped["Spend"]]  
families_income_less_than_spend是空集合因此推定不存在入不敷出的  
家庭。
```


Q3. Are there any single-parent families, where only one Adult is present? Are there any childless families? How do you discern this?

1. Only one Adult is present

	Family	Member	Income	Spend
8	family3	Adult1	2301931	807835
20	family7	Adult1	2815513	1295014
21	family7	Child1	0	3612
39	family14	Adult1	998732	248730
40	family14	Child1	0	9006
...	...	...	...	...
265	family94	Child1	0	16485
269	family96	Adult1	325062	117236
270	family96	Child1	0	7530
271	family96	Child2	0	11188
276	family99	Adult1	1827150	493578

73 rows × 4 columns

2. Childless families are also present

	Family	Member	Income	Spend
0	family1	Adult1	2376330	1119433
1	family1	Adult2	130268	37337
2	family1	Adult3	2254489	972327
3	family2	Adult1	2292355	649806
4	family2	Adult2	298167	100723
...	...	...	...	...
274	family98	Adult1	1445541	587647
275	family98	Adult2	1573068	444308
276	family99	Adult1	1827150	493578
277	family100	Adult1	751899	172111
278	family100	Adult2	279747	86303

188 rows × 4 columns

3. 我是以下列程式篩選出來的

(1) 將 "single_Adult" 篩選出來

```
# 將 "Adult" 篩選出來adults =
df[df["Member"].str.contains("Adult")]
```

```
# 計算每個 family 中 Adult 的數量
adult_counts = adults.groupby("Family").size()

# 篩選出只有一個 Adult 的 family
single_adult_families = adult_counts[adult_counts == 1].index

(2) 將 "no_Child" 篩選出來
no_childs = df[~df["Member"].str.contains("Child")]
```

Q4. Do you suspect any errors within this dataset? Examples may include negative figures, missing or duplicate data, etc. Why?

想法一:將df中缺失值、重複值、負數值分別取sum(),可以知道出現錯誤的資料數量。

想法二:將df中缺失值、重複值、負數值分別存入新建立的df中,相較於想法一,能一目瞭然看到哪筆資料錯誤。

不過兩種檢查方式在這裡都沒有看到錯誤。

Part 2: Questions Related to the Use of AI Tools:

A. Did you use AI tools to assist in completing Q1 to Q4?

NO 我僅有使用 AI當字典查詢我想到的篩選方式該如何化為程式碼。